

# TruthGPU: A Parity-Gated, Cloud-to-Edge Full Stack for Accelerated and Robust LLM Safety and Hallucination Verification

AEGIS Research Team, Yataav Inc.<sup>1</sup>

<sup>1</sup>AEGIS Research, Yataav Inc. — AI Safety & Reliability Verification Infrastructure

Technical Report — v1 (2026). **Pre-publication draft; results on internal/synthetic and zero-overlap held-out data unless noted.**

## Abstract

Deploying large language models (LLMs) at scale requires two verification layers—an input-side *security guardrail* (against prompt injection, jailbreaks, and harmful content) and an output-side *hallucination verifier*—each of which is itself a neural-network inference and therefore a latency and cost bottleneck. We present **TruthGPU**, a full-stack verification infrastructure that (i) serves both engines from a *single* GPU service with shared models and acceleration; (ii) *guarantees* that acceleration does not alter the safety verdict via an automated *parity gate*; (iii) resolves the robustness–precision tradeoff of safety classifiers through a data-centric closed loop rather than post-hoc calibration; (iv) augments natural-language inference (NLI) hallucination detection with a context-category-gated numeric cross-verification layer; and (v) distills the same policy and taxonomy onto a family of on-device security semi-conductors for offline, data-sovereign operation. On an NVIDIA H200 we measure up to  $11.4\times$  NLI and  $4.57\times$  end-to-end acceleration with *zero* verdict change; we raise out-of-distribution adversarial robustness from 0.759 to 0.956 macro-F1 while *simultaneously* reducing the benign false-positive rate from 0.05 to 0.025; and we improve numeric-hallucination recall from 0.667 to 0.889 at a zero false-hallucination rate. We are deliberate about scope: speed, cost, and verdict-preservation are measured, whereas the *external* generalization of detection quality and on-silicon edge latency remain open and are reported as such.

## 1 Introduction

LLM deployments face two distinct failure modes. *Security*: adversarial inputs (prompt injection, jailbreaks, encoding/obfuscation evasion) can subvert a model into producing prohibited output. *Hallucination*: fluent but

factually wrong answers, particularly numeric and relational errors, erode trust. Both are mitigated by a *verification layer*, but that layer is another neural-network inference, adding latency and cost—a bottleneck that worsens as guardrails grow more capable.

A second, subtler problem is *evaluation optimism*. Safety classifiers are typically trained and scored on data from the same synthetic generator, yielding self-graded accuracies near 0.99 that collapse to  $\sim 0.76$  on genuinely out-of-distribution (OOD) obfuscated attacks. A central thesis of this work is that this gap is a *data* problem, not a thresholding problem, and is therefore not repairable by post-hoc calibration.

A third requirement is *sovereignty*: many deployments (automotive, medical, defense, industrial) must verify offline, without sending data off-device. This motivates pushing the same verification policy onto edge silicon.

**Contributions.** We make six contributions, integrated into one system:

- A Parity-gated acceleration:** fused GPU kernels for verification post-processing, coupled to an automated gate that enforces *verdict-label* equivalence (not merely numerical closeness) between the accelerated and reference paths before any deployment (§4.1).
- B Numeric $\oplus$ NLI hallucination detection:** a numeric cross-verification layer, gated by semantic-category compatibility and unit normalization, placed *before* NLI (§4.2).
- C Adversarial + targeted hard-negative training:** a closed loop that reveals optimism with a zero-overlap held-out probe, raises robustness by LLM-generated adversarial augmentation, then recovers precision by LLM-generated *targeted* hard-

negatives of the regressed benign distribution (§4.3).

**D Multi-signal fusion:** union of the ML classifier with deterministic rules designed to be absent from the benign distribution, gated for non-harm (§4.4).

**E Safety-constrained selective prediction:** operating-point selection under a hard constraint that *no* attack is auto-allowed (§4.5).

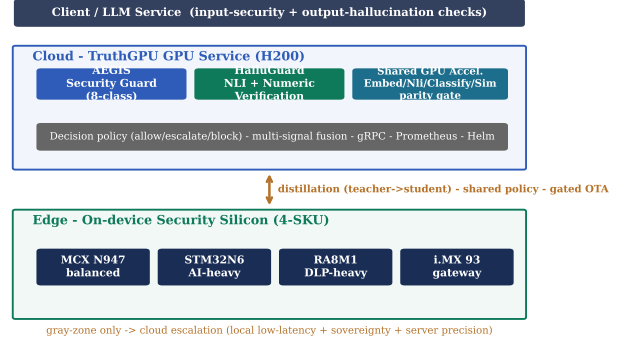
**F Cloud-to-edge unification:** distillation of the same taxonomy, policy, and interface contract onto a 4-SKU silicon family with gated over-the-air (OTA) updates (§4.6).

## 2 Related Work

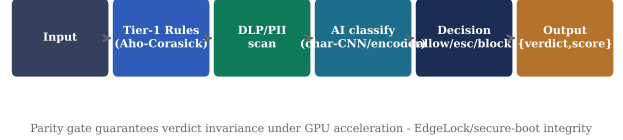
**Guardrails.** Moderation/guard classifiers (e.g., Llama Guard [3]) detect unsafe content; we add a verdict-preserving acceleration gate and a data-centric robustness loop. **Hallucination.** NLI- and retrieval-based fact verification (e.g., FEVER [6]) underpins output checking; we add category-gated numeric verification as a pre-NLI layer. **Acceleration.** Kernel fusion and quantization (e.g., TensorRT [5]) speed inference; we contribute the enforcement of *verdict* equivalence as a deployment gate. **Adversarial training** and **hard-negative mining** [4] improve robustness; we combine them in a leakage-verified closed loop that also restores precision. **Selective prediction** [1] trades coverage for accuracy; we add an asymmetric no-attack-leak constraint. **Distillation** [2] and edge inference enable on-device models; we share a single taxonomy/policy/contract across heterogeneous compute with gated OTA.

## 3 System Architecture

TruthGPU exposes a single GPU service (Fig. 1) offering four RPCs (EMBED, NLI, CLASSIFY, SIMILARITY) shared by both engines, so models are loaded once. The security engine (AEGIS) performs 8-class multilabel guarding; the hallucination engine (HalluGuard) performs NLI plus numeric verification. A common decision policy, multi-signal fusion, Prometheus observability, and a Helm deployment chart complete the service. Each input flows through a layered pipeline (Fig. 2): deterministic rules and PII/DLP scanning resolve clear cases on CPU; ambiguous cases invoke the accelerated classifier; a decision policy emits {allow, escalate, block}.



**Figure 1:** TruthGPU full stack. One GPU service hosts both engines and shared acceleration; the same taxonomy/policy is distilled onto a 4-SKU edge silicon family, which escalates only gray-zone inputs to the cloud.



**Figure 2:** Layered verification pipeline. Cheap deterministic stages short-circuit clear cases; the parity gate guarantees verdict invariance under acceleration.

## 4 Methods

### 4.1 Parity-Gated Acceleration (A)

The post-processing of a verification network—class-order permutation, normalization, threshold comparison, and verdict computation—is executed within a *single* GPU kernel invocation, eliminating host-device round-trips. Let  $z(x) \in \mathbb{R}^K$  be the logits,  $\pi$  a model-specific class permutation,  $\sigma$  the normalizer, and the discrete verdict

$$v(x) = \arg \max_k [\sigma(\pi z(x))]_k, \quad c(x) = \max_k [\sigma(\pi z(x))]_k. \quad (1)$$

Writing  $v_{\text{acc}}, c_{\text{acc}}$  for the accelerated kernel and  $v_{\text{ref}}, c_{\text{ref}}$  for the CPU reference, the *parity* predicate over a verification set  $V$  is

$$G_{\text{par}} = \left[ \sum_{x \in V} \mathbf{1}[v_{\text{acc}}(x) \neq v_{\text{ref}}(x)] = 0 \right] \wedge \left[ \max_{x \in V} |c_{\text{acc}}(x) - c_{\text{ref}}(x)| \right] \quad (2)$$

The accelerated path is deployed iff  $G_{\text{par}}$  holds *and* every quality-regression metric  $m \in \mathcal{M}$  (attack TPR, benign FPR, OOD robustness, fusion) meets its threshold  $\tau_m$  (Alg. 1). The novelty is coupling single-kernel transfer elimination with *verdict-level* equivalence (Eq. 2) enforced as an automated gate.

---

**Algorithm 1** Parity-gated deployment

---

**Require:** accelerated kernel  $f_{\text{acc}}$ , reference  $f_{\text{ref}}$ , set  $V$ , thresholds  $\varepsilon, \{\tau_m\}$

- 1:  $\text{mis} \leftarrow \sum_{x \in V} \mathbf{1}[v_{\text{acc}}(x) \neq v_{\text{ref}}(x)]; \delta \leftarrow \max_x |c_{\text{acc}}(x) - c_{\text{ref}}(x)|$
- 2: **if**  $\text{mis} > 0$  **or**  $\delta > \varepsilon$  **then**
- 3:     **return** BLOCK (parity fail)
- 4: **end if**
- 5: **for**  $m \in \mathcal{M}$  **do**
- 6:     **if**  $\text{metric}_m(f_{\text{acc}}) < \tau_m$  **then**
- 7:         **return** BLOCK (regression)
- 8:     **end if**
- 9: **end for**
- 10: **return** DEPLOY

---



---

**Algorithm 2** Numeric $\oplus$ NLI layered verdict

---

**Require:** claim  $q$ , evidence  $e$

- 1:  $F_c \leftarrow \text{EXTRACTNUMERIC}(q); F_e \leftarrow \text{EXTRACTNUMERIC}(e)$   $\triangleright$  ambiguous units need explicit marker
- 2: **for**  $(f_c, f_e) \in F_c \times F_e$  with  $\text{Comparable}(f_c, f_e)$  **do**
- 3:     **if**  $|\eta(\nu_c) - \eta(\nu_e)| > \theta(u_c)$  **then**
- 4:         **return** (CONTRADICTED,  $c=0.92$ )
- 5:     **end if**
- 6: **end for**
- 7: **return** NLI( $q, e$ )

---

## 4.2 Numeric $\oplus$ NLI Hallucination Detection (B)

We extract numeric facts  $f = (\nu, u, \gamma)$  (value, unit, context keyword) from claim and evidence; for ambiguous units a number is extracted only with an explicit marker (e.g. °C), excluding bare degree signs. Let  $\kappa(\gamma)$  map a context to a semantic category and  $\text{cmp}(u_c, u_e)$  test unit compatibility. A claim fact  $f_c$  and evidence fact  $f_e$  are comparable iff

$$\text{Comparable}(f_c, f_e) = [\kappa(\gamma_c) = \kappa(\gamma_e)] \wedge \text{cmp}(u_c, u_e), \quad (3)$$

which blocks cross-domain mismatch. A contradiction is declared when, after unit normalization  $\eta$ , the gap exceeds a per-unit tolerance  $\theta(u)$  (absolute tolerance applied *before* the zero-value rule):

$$\text{Contra}(f_c, f_e) = \text{Comparable}(f_c, f_e) \wedge [|\eta(\nu_c) - \eta(\nu_e)| > \theta(u_c)] \quad (4)$$

The layered verdict applies the numeric layer *before* NLI (Alg. 2):

$$\text{verdict} = \begin{cases} \text{CONTRADICTED}, & \exists (f_c, f_e) : \text{Contra}(f_c, f_e), \\ \text{NLI}(\text{claim}, \text{evidence}), & \text{otherwise.} \end{cases} \quad (5)$$

---

**Algorithm 3** Adversarial + targeted hard-negative loop

---

**Require:**  $D$ , held-out  $H$  ( $H \cap D = \emptyset$ ), benign  $B$ , gates  $R_{\min}, P_{\max}$

- 1:  $\theta \leftarrow \text{TRAIN}(D)$   $\triangleright$  reveal:  $R(\theta) \ll \text{in-sample}$
- 2:  $A \leftarrow \text{LLMADVERSARIAL}(); D \leftarrow D \cup A; \theta \leftarrow \text{TRAIN}(D)$   $\triangleright R \uparrow, P \uparrow(\text{worse})$
- 3:  $B^* \leftarrow \{x \in B : \hat{y}_\theta(x) = \text{unsafe}\}$
- 4:  $N \leftarrow \text{LLMGENERATE}(B^*) \setminus (H \cup \text{eval}); D \leftarrow D \cup N; \theta \leftarrow \text{TRAIN}(D)$   $\triangleright P$  restored
- 5: **if**  $R(\theta) \geq R_{\min} \wedge P(\theta) \leq P_{\max}$  **then**
- 6:     **return**  $\theta$
- 7: **else**
- 8:     **return** BLOCK
- 9: **end if**

---

## 4.3 Adversarial + Targeted Hard-Negative Training (C)

Let  $D$  be training data,  $H$  a *structure-independent* held-out probe with verified  $H \cap D = \emptyset$ ,  $B$  a benign set. Define robustness  $R(\theta) = \text{macroF1}_H(\theta)$  and false-positive rate  $P(\theta) = \text{FPR}_B(\theta)$ . The closed loop (Alg. 3) first exposes the gap  $R(\theta_1) \ll \text{in-sample F1}$ , then augments with LLM adversarial samples  $A$  (whitespace, zero-width, Jamo decomposition, code-mixing) to raise  $R$ , then *targets* the regressed benign distribution: it identifies  $B^* = \{x \in B : \hat{y}_\theta(x) = \text{unsafe}\}$  and adds LLM-generated hard-negatives  $N \sim \mathcal{G}(B^*)$  with  $N \cap (H \cup \text{eval}) = \emptyset$  to restore precision. The dual gate deploys iff

$$R(\theta) \geq R_{\min} \wedge P(\theta) \leq P_{\max}. \quad (6)$$

We further show post-hoc calibration  $\min_{t,T} P$  cannot raise  $R$  (it is constant in  $t, T$  on  $H$ ), evidencing the data-lever thesis. For the edge student, distillation minimizes

$$\mathcal{L} = \alpha \text{CE}(y, \sigma(z_s)) + (1 - \alpha) T^2 \text{KL}(\sigma(z_t/T) \parallel \sigma(z_s/T)). \quad (7)$$

## 4.4 Multi-Signal Fusion (D)

With ML score  $s(x)$ , threshold  $\theta$ , and a rule set  $\mathcal{R}$  *designed so that*  $\Pr_{x \sim B}[\exists r \in \mathcal{R} : r(x) = 1] \approx 0$ , the fused verdict is the union

$$\text{unsafe}(x) = \mathbf{1}[s(x) \geq \theta] \vee \bigvee_{r \in \mathcal{R}} r(x). \quad (8)$$

$\Delta \text{FP}_B(\mathcal{R}) = 0 \wedge \text{TPR}_{\text{fused}} \geq \text{TPR}_{\text{ML}}$ . benign false positives and does not lower detection:

## 4.5 Safety-Constrained Selective Prediction (E)

Scores route to auto-allow ( $s < t_\ell$ ), auto-block ( $s \geq t_h$ ), or review (otherwise). With coverage  $C = \Pr[s < t_\ell \vee$

$s \geq t_h]$  and auto-accuracy  $A$ , we select

$$(t_\ell^*, t_h^*) = \arg \max_{t_\ell \leq t_h} C(t_\ell, t_h) \text{ s.t. } A \geq A^* \wedge \forall x \in \mathcal{A} : s(x) \geq t_\ell, \quad (9)$$

where  $\mathcal{A}$  is the attack set; the second constraint forbids *any* auto-allowed attack—an asymmetric-risk condition absent from standard selective prediction. Bands violating it (e.g.  $[0.5, 0.5]$ ) are excluded regardless of accuracy.

## 4.6 Cloud-to-Edge Unification (F)

A server-side accelerated verifier (teacher) and a device-side distilled verifier (student, INT8, offline-capable) share an identical verdict label space, interface contract, and decision policy, so verdict semantics are portable across tiers. Retrained models propagate by gated OTA (regression gate before hot-swap). The portable core (rules, DLP, preprocessing, decision) is silicon-independent; per-chip HAL backends implement NPU inference, I/O, and the hardware root-of-trust (Edge-Lock/RSIP/secure boot).

## 5 Experimental Setup

**Hardware.** NVIDIA H200 (143 GB), CUDA 13. **Models.** Multilingual NLI (mDeBERTa-v3 XNLI) and an 8-class guard encoder. **Data.** (i) a synthetic red-team set (in-distribution); (ii) a *zero-overlap* held-out OOD probe with obfuscated/multilingual attacks; (iii) benign sets (in-distribution, OOD, and sensitive-but-legitimate). **Edge.** distilled char-CNN/MLP students; three silicon backends (NXP MCX N947, ST STM32N6, Renesas RA8M1) plus an i.MX 93 gateway tier. **Honesty.** We separate *measured*<sup>†</sup> results (latency, parity, footprint, and metrics on our own held-out data) from *design/estimate*<sup>†</sup> values (on-silicon edge latency) and flag where evaluation is self-graded synthetic vs. zero-overlap held-out.

## 6 Results

### 6.1 Acceleration and Verdict Preservation

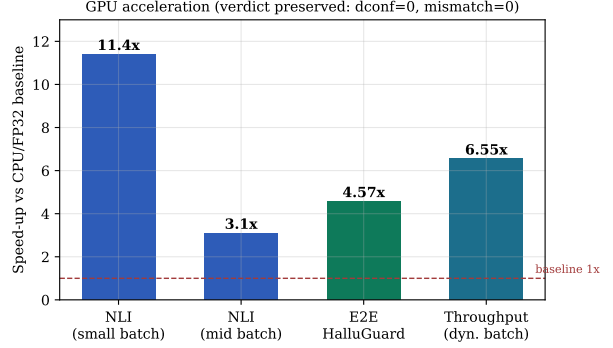
Fig. 3 and Table 1: up to 11.4× NLI (small batch), 4.57× end-to-end, 6.55× throughput, ~3× TCO reduction; the fused kernel exhibits  $1.8 \times 10^{-7}$  numerical error with *zero* top-1 mismatch and zero confidence delta—i.e., faster with an identical safety verdict.

### 6.2 Robustness–Precision (Data Closed-Loop)

Fig. 4 and Table 2: adversarial augmentation lifts OOD robustness 0.759→0.978 but degrades benign precision

**Table 1:** Parity-gated acceleration on H200 (verdict preserved).<sup>†</sup>

| Metric                           | Value                    |
|----------------------------------|--------------------------|
| NLI inference (small batch)      | 11.4×                    |
| NLI inference (mid batch)        | 3.1×                     |
| End-to-end HalluGuard            | 4.57×                    |
| Throughput (dynamic batching)    | 6.55×                    |
| Numerical error / top-1 mismatch | $1.8 \times 10^{-7}$ / 0 |
| Confidence delta (acc. vs. ref.) | 0                        |



**Figure 3:** Speed-ups vs. baseline; the parity gate certifies  $\Delta\text{conf} = 0$  and mismatch=0.

(FPR 0.05→0.15); threshold calibration fails to recover it; targeted hard-negatives restore FPR to 0.025 while holding robustness at 0.956. Held-out/training overlap is verified to be zero, supporting genuine generalization.

### 6.3 Hallucination Detection

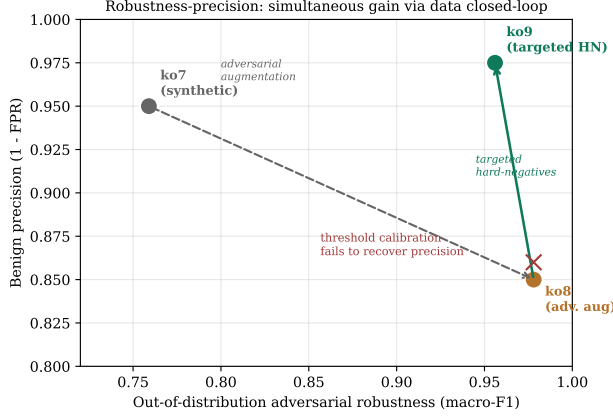
Fig. 5: numeric⊕NLI layering raises contradiction recall 0.667→0.778 (financial)→0.889 (temperature) at a *zero* false-hallucination rate, with no cross-domain or angle/coordinate mis-extraction.

### 6.4 Fusion, Selective Prediction, Edge

Multi-signal fusion raises adversarial macro-F1 0.956→0.978 with *zero* added false positives (Fig. 6a). Safety-constrained selective prediction at band  $[0.1, 0.9]$  yields 100% auto-accuracy at 99.3% coverage (0.7% review) with zero auto-allowed attacks; the  $[0.5, 0.5]$  band is excluded for leaking two attacks. The edge student is a 0.113 MB INT8 char-CNN with F1 0.998 and FNR 0 on its (binary, in-distribution) test set (Fig. 6b); all three silicon backends link against the shared core, and a Rust runtime E2E confirms the live GPU path (engine = GuardEncoder; attack 0.9996 / benign 0.0004).

**Table 2:** Guard model evolution (zero-overlap held-out OOD).<sup>†</sup>

| Stage             | OOD macro-F1 | FPR          | Lever                 |
|-------------------|--------------|--------------|-----------------------|
| ko7 (synthetic)   | 0.759        | 0.05         | large benign          |
| ko8 (adv. aug)    | <b>0.978</b> | 0.15         | robustness $\uparrow$ |
| threshold cal.    | —            | fails        | post-hoc              |
| ko9 (targeted HN) | 0.956        | <b>0.025</b> | robust + precise      |



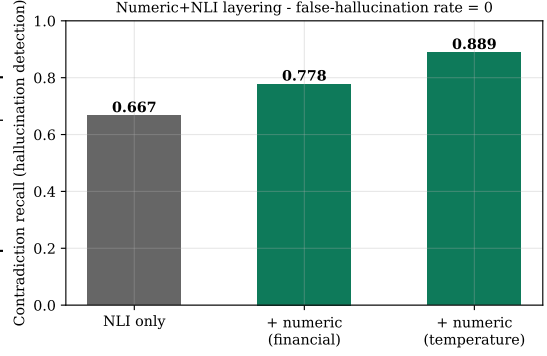
**Figure 4:** Robustness and precision improved *simultaneously* by the data closed-loop; post-hoc calibration cannot move along the robustness axis.

## 7 Discussion and Limitations

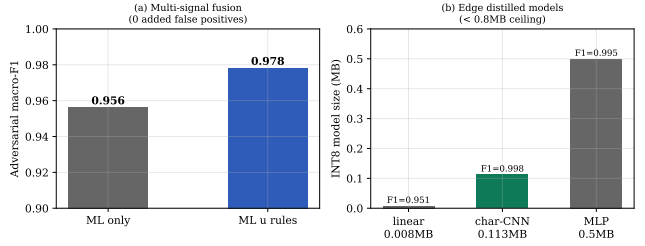
We are explicit about what is *not* yet established. (1) **External generalization.** Detection-quality metrics use our own synthetic and zero-overlap held-out data; real-customer traffic is unvalidated. The headline honesty result—that self-graded 0.99 becomes 0.76 OOD—is itself a warning that internal numbers are upper bounds. (2) **Edge silicon.** On-device latency/power are design estimates<sup>‡</sup>; real measurement awaits board bring-up and NPU compilation (Vela/eIQ/Hailo op coverage, notably the embedding **Gather**). (3) **Mid-size edge encoders** (5–80 MB) are a design option, not yet trained/measured. (4) **Selective-prediction** 100% is on a small, well-separated set and should not be read as a universal guarantee. We report these as open items rather than smoothing them over; the system’s gates are designed precisely to prevent such optimism from shipping.

## 8 Conclusion

TruthGPU unifies LLM security and hallucination verification in one accelerated GPU service that provably preserves the safety verdict, attains robustness and precision *together* through a data-centric loop rather than post-hoc tuning, and extends the same policy to



**Figure 5:** Numeric cross-verification recovers contradictions NLI misses; false-hallucination rate = 0.



**Figure 6:** (a) Fusion adds recall at zero added FP. (b) Distilled edge models fit the < 0.8 MB on-device ceiling.

data-sovereign edge silicon. Speed, cost, and verdict-preservation are measured; external detection-quality generalization and on-silicon latency are the next milestones.

## A Dataset Statistics

Table 3 summarizes the corpora. Cloud benign data mixes real-world, LLM-generated, augmented, and *sensitive-but-legitimate* (security-query) sets; attacks are synthetically generated and capped at 3000 for gate evaluation, while the OOD probe is structure-independent with verified zero overlap with training data (§4.3). The edge distillation set is a binary Safe-vs-Attack corpus (hash  $n$ -grams {3, 5}, dim 8192).

## B Hyperparameters

## C Additional Ablations

**Fusion (D).** The  $ML \cup rule$  union raises adversarial macro-F1 with zero added false positives on each benign split (Table 5). **Calibration.** Post-hoc threshold/temperature tuning on ko8 leaves OOD robustness unchanged and does not recover FPR; only data (ko9) does (§4.3), evidencing the data-lever thesis. **Selective (E).** A band sweep (Table 6) shows narrow bands leak

**Table 3:** Dataset statistics (counts).

| Corpus             | Size         | Role                       |
|--------------------|--------------|----------------------------|
| benign_real        | 2000         | real-world benign          |
| benign_llm (+eval) | 3328 (587)   | LLM-generated benign       |
| benign_aug         | 1800         | augmented benign           |
| benign_eval        | 400          | held-out benign            |
| security_benign    | 219          | sensitive-but-legitimate   |
| red-team (attacks) | $\leq 3000$  | synthetic, in-distribution |
| OOD probe          | held-out     | structure-indep., overlap  |
| edge total         | 5252         | binary Safe-vs-Attack      |
| train / val / test | 3673/784/795 | attack rate 0.79           |

**Table 4:** Training configurations.

| Component                                                      | Setting                                                  |
|----------------------------------------------------------------|----------------------------------------------------------|
| <i>Guard fine-tune (8-class, multi-label)</i>                  |                                                          |
| base model                                                     | mDeBERTa-v3-base                                         |
| objective                                                      | multi-label BCE                                          |
| epochs / batch / lr                                            | 4-6 / 16 / $2 \times 10^{-5}$                            |
| max length / optim / seed                                      | 128 / AdamW / 42                                         |
| <i>Distillation (teacher <math>\rightarrow</math> student)</i> |                                                          |
| teacher                                                        | mDeBERTa-v3 (278M)                                       |
| loss                                                           | $(1-\alpha) \text{BCE}_{\text{hard}} + \alpha \text{KD}$ |
| KD / T / $\alpha$                                              | $\text{BCE}(\text{student}, \text{teacher\_probs}) / 1$  |
| epochs / lr                                                    | 5 / $3 \times 10^{-5}$                                   |
| <i>NLI (hallucination)</i>                                     |                                                          |
| model                                                          | mDeBERTa-v3 XNLI (2mil7)                                 |
| label remap                                                    | 2, 0, 1                                                  |
| <i>Edge char-CNN student</i>                                   |                                                          |
| input / vocab                                                  | char ids / 878                                           |
| quant / size                                                   | INT8 / 0.113 MB                                          |

**Table 6:** Selective-prediction (E) band sweep.

| $[t_\ell, t_h]$ | Auto-acc.   | Coverage     | Attack leak  |
|-----------------|-------------|--------------|--------------|
| [0.5, 0.5]      | high        | 100%         | 2 (excluded) |
| [0.2, 0.8]      | 100%        | $\sim 98\%$  | 0            |
| [0.1, 0.9]      | <b>100%</b> | <b>99.3%</b> | <b>0</b>     |

**Table 7:** Numeric $\oplus$ NLI (B): contradiction recall.

| Config                  | Recall       | False-halluc. |
|-------------------------|--------------|---------------|
| NLI only                | 0.667        | 0             |
| + numeric (financial)   | 0.778        | 0             |
| + numeric (temperature) | <b>0.889</b> | 0             |

attacks; [0.1, 0.9] attains 100% auto-accuracy at 99.3% coverage with zero leakage. **Numeric (B).** Adding numeric layers to NLI lifts contradiction recall monotonically at zero false-hallucination (Table 7). **Edge.** Footprint/accuracy across student capacities (Table 8); all reach FNR = 0 on the test split, with per-technique recall 1.0 on social-engineering, prompt-injection, and encoding/script families.

**Table 5:** Multi-signal fusion (D): added false positives by benign split.

| Config          | Adv. macro-F1 | $\Delta$ FP (in/OOD/sens.) |
|-----------------|---------------|----------------------------|
| ML only         | 0.956         | —                          |
| ML $\cup$ rules | <b>0.978</b>  | 0/0/0                      |

## References

- [1] Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *NeurIPS*, 2017.

**Table 8:** Edge student capacity (INT8).

| Model    | Size (MB) | F1           | FNR |
|----------|-----------|--------------|-----|
| linear   | 0.008     | 0.951        | 0   |
| char-CNN | 0.113     | <b>0.998</b> | 0   |
| MLP      | 0.500     | 0.995        | 0   |